

Lemma 1. *If R is a UFD, and $r = \prod_{i=1}^n p_i$ is some irreducible decomposition, then*

$$d \mid r \iff d = u \prod_{j \in S \subseteq [n]} p_j \text{ for some subset } S \subseteq [n] \text{ and some unit } u \in R.$$

Proof. Let $d = d_1 \cdots d_m$ be some irreducible decomposition, which is unique up to associates since R is a UFD. ($\implies$) If $d \mid r$, $\exists q \in R$ such that $r = dq$. and there exists some irreducible decomposition $q = q_1 \cdots q_t$.

$r = \prod_{i=1}^n p_i = dq = (d_1 \cdots d_m)(q_1 \cdots q_t)$. Recall that since R is a UFD, and r decomposes into n many p_i 's, $r = dq \implies dq$ must decompose into n many d_i 's and q_i 's, that is $t + m = n$. So reindex the q_i 's so that $q_i = q_{i+m}$ for all $i \in [t]$ and we obtain $r = p_1 \cdots p_m q_{m+1} \cdots q_n$. Since R is a UFD, there exists a permutation $\sigma \in S_n$ such that for some units $u_1, \dots, u_n \in R$,

$$d_i = u_i p_{\sigma(i)}, \forall i \in [m] \text{ and } q_i = u_i p_{\sigma(i)}, \forall m < i \leq n.$$

That is, if $S = \sigma([m])$, since R is commutative,

$$d = \left(\prod_{i=1}^m u_i \right) \left(\prod_{j \in S \subseteq [n]} p_j \right) = u \prod_{j \in S \subseteq [n]} p_j \text{ where } u = \prod_{i=1}^m u_i.$$

($\impliedby$) On the other hand, if $d = u \prod_{j \in S \subseteq [n]} p_j$ for some unit $u \in R$, we have that:

$$(u^{-1} \prod_{k \in [n] \setminus S} p_k) d = (u^{-1} \prod_{k \in [n] \setminus S} p_k) (u \prod_{j \in S \subseteq [n]} p_j) = \prod_{i \in [n]} p_i = r \implies d \mid r.$$

□

Lemma 2. *If R is Noetherian containing a prime ideal P , then R/P is also Noetherian.*

Proof. Let $\pi : R \rightarrow R/P$ be the canonical homomorphic projection, and I_P any ideal of R/P . π is surjective, so $I = \pi^{-1}(I_P) \subseteq R$ must be a finitely generated ideal since R is Noetherian. That is, $I = \pi^{-1}(I_P) = \langle i_1, \dots, i_n \rangle \implies I_P = \pi(I) = \langle \pi(i_1), \dots, \pi(i_n) \rangle$. Therefore, R/P is Noetherian.

□

Lemma 3. *Let R be a Noetherian UFD, and let P be a prime ideal of R . Then there exist some prime elements $q_1, \dots, q_n \in P$ such that $P = \langle q_1, \dots, q_n \rangle$.*

Proof. Since R is Noetherian, P is finitely generated. So there exist some $a_1, \dots, a_n \in P$ such that $P = \langle a_1, \dots, a_n \rangle$. Since R is a UFD, for each $i \in [n]$, $a_i = \prod_{j=1}^{m_i} p_{(i)_j}$ for some irreducible and therefore prime $p_{(i)_j}$'s. Well, $\prod_{j=1}^{m_i} p_{(i)_j} \in P \implies q_i = p_{(i)_k} \in P$ for some $k \in [m_i]$. So then $a_i = q_i \prod_{\substack{j \in [m_i] \\ p_{(i)_j} \neq q_i}} p_{(i)_j} \implies P = \langle q_1, \dots, q_n \rangle$.

□

Lemma 4 (Problem 7 from HW 3). *Let R be a commutative ring, and let $I \subseteq R$ be a proper ideal. Prove that the set of prime ideals containing I has minimal elements. We call such a minimal element a minimal prime ideal over I .*

Proof. Let S be the set of prime ideals of R that contain I . In **Homework 2** we proved that the set of all proper ideals containing I is a poset with set inclusion. So then $(S \setminus \{R\}, \subseteq)$ is an induced subposet of that poset. Obviously, $P \subseteq R$, $\forall P \in S \setminus \{R\}$ and so $(S, \subseteq)$ is also a poset. Now consider any chain C in S .

Let $L_C = \bigcap_{P \in C} P$. Obviously, $L_C \subseteq C, \forall P \in C \implies I \subseteq L_C$. Next, $\forall a, b \in L_C, a, b \in P, \forall P \in C \implies a - b \in P, \forall P \in C$ since they are all additive subgroups of R . So $L_C \leq_+ R$. Also, $\forall r \in R, \forall a \in L_C, ra \in P, \forall P \in C \implies ra = ar \in L_C$. So L_C is an ideal of R containing I . Finally, if $ab \in L_C$, then $ab \in P, \forall P \in C \implies a$ or $b \in P, \forall P \in C \implies a$ or $b \in L_C$. So L_C is a prime ideal of R containing I , and a lower bound for C in S . So every chain in S has a lowerbound.

So then in the dual poset S^* every chain has an upperbound in S , and by Zorn's Lemma, there exists a maximal element in S^* . That is, there exists a minimal element in S . □

Lemma 5. *A prime ideal P of a UFD R has height at most 1 if and only if P is principal.*

Proof. ($\implies$) If P has height 1, then there is no prime ideal strictly between P and $\{0\}$. If $P = \{0\} = \langle 0 \rangle$, then it's principal. If P is non-zero, suppose it is not principal. So P contains at least two non-zero coprime generators x and y . R is a UFD, so x has a prime decomposition $x = p_1 \cdots p_n$. Therefore, since P is prime, some p_j belongs to P . So $\langle p_i \rangle$ is prime and $0 \subset \langle p_i \rangle \subseteq P \implies P$ is principal or $\{0\} \subset \langle p_j \rangle \subset P \implies \text{ht}(P) > 1$, both contradictions. Thus, P is principal.

($\impliedby$) If $P = \{0\} = \langle 0 \rangle$, it obviously has height $0 \leq 1$. If P is principal and prime, then $P = \langle p \rangle$ for some prime element $p \in R \setminus \{0\}$. Suppose there exists some prime ideal Q strictly between $\{0\}$ and P . Then there exists some non-zero $q \in Q$ which has a prime decomposition $q = q_1 \cdots q_n$ since R is a UFD and since Q is prime, some q_j belongs to Q . Recall that all primes are irreducible and vice versa, so then since $q_j \in Q \subset P \implies \exists r \in R$ s.t. $q_j = rp$, r or p is a unit. No primes are units, so r is a unit. Well, then $q_j = rp \in Q \implies (p \in Q \implies P = \langle p \rangle \subseteq Q)$ or $(r \in Q \implies rr^{-1} = 1 \in Q \implies Q = R \subset P)$, which are both contradictions. Thus, there is no prime ideal strictly between $\{0\}$ and P , so $\text{ht}(P) = 1$. □

Lemma 6. *If R is Noetherian domain, then a principal prime ideal P of R has height at most 1*

Proof. Let R be a Noetherian integral domain. If $P = \{0\} = \langle 0 \rangle$, it has height $0 \leq 1$. If P is prime, principal, and non-zero, then $P = \langle p \rangle$ for some prime element $p \in R \setminus \{0\}$. Suppose there exists a prime ideal Q such that $\{0\} \subset Q \subset P$. Then $\exists q \in Q \setminus \{0\} \subset P \implies \exists r_0 \in R$ such that $q = r_0 p$. Since Q is prime we have that $r_0 \in Q$ or $p \in Q$. but p can't belong to Q since $Q \subset P = \langle p \rangle$, so $r_0 \in Q$.

We can define a sequence $(r_i)_{i \in \mathbb{N}}$ in R recursively by repeating the same process indefinitely

$$r_i p^{i+1} \in Q \implies r_i \in Q \implies r_i = r_{i+1} p \text{ for some } r_{i+1} \in R \implies q = r_i p^{i+1}, \forall i \in \mathbb{N}.$$

Now, since $r_i = r_{i+1} p, \forall i \in \mathbb{N}$, we obtain an ascending chain of ideals

$$\langle r_0 \rangle \subseteq \langle r_1 \rangle \subseteq \cdots \subseteq \langle r_i \rangle \subseteq \cdots$$

which must stabilize at some $i = n \in \mathbb{Z}^+$ since R is Noetherian. We already have that $r_n = r_{n+1} p$ and now $\langle r_n \rangle = \langle r_{n+1} \rangle \implies r_{n+1} = s r_n$ for some $s \in R$. So then $r_n = r_{n+1} p = (s r_n) p \implies r_n - s r_n p = r_n(1 - s p) = 0 \implies 1 - s p = 0$. But then since $q = s r_n p^{n+1} \neq 0$ and R is an integral domain, s is non zero and p is a prime unit. The very thing is impossible.

Therefore there is no prime ideal strictly between $\{0\}$ and P , and $\text{ht}(P) \leq 1$.

□

Corollary 7 (Corollary of Homework 3 Problem 9). *If R is a commutative ring and P is a prime ideal of R , then the set of all prime ideals of R contained in P is in bijection with the set of all prime ideals of R_P via $Q \longleftrightarrow QR_P$.*

Proof. Since P is prime and R is commutative, $R_P = \{\frac{r}{q} \mid r \in R, q \in R \setminus P\}$ is a local ring with maximal ideal PR_P . Additionally, by **Homework 3**, all ideals of R_P are of the form $IR_P = \{\frac{i}{q} \in R_P \mid i \in I\}$ for some $I \subseteq R$, and $I \not\subseteq P \implies IR_P = R_P$. We show an ideal $Q \subset P$ is prime if and only if QR_P is prime.

($\implies$) Suppose $Q \subset P$ is prime. If $(\frac{a}{r})(\frac{b}{s}) \in QR_P$, then $\frac{ab}{rs} \in QR_P \implies ab \in Q \implies a \in Q$ or $b \in Q \implies \frac{a}{r} \in QR_P$ or $\frac{b}{s} \in QR_P$. So $Q \subset R$ is prime $\implies QR_P$ is prime.

($\impliedby$) On the other hand, if QR_P is prime, $ab \in Q \implies \frac{ab}{1} = (\frac{a}{1})(\frac{b}{1}) \in QR_P \implies \frac{a}{1} \in QR_P$ or $\frac{b}{1} \in QR_P \implies a \in Q$ or $b \in Q$. So QR_P is prime $\implies Q \subset P$ is prime.

So we see that QR_P is prime $\iff Q \subset P$ is prime and immediately the mapping ϕ from prime ideals Q properly contained in P to prime ideals of R_P defined by $\phi(Q) = QR_P$ is a surjection. Now, if $Q_1 \neq Q_2$ but both are prime ideals properly contained in P , consider $Q_1 R_P$ and $Q_2 R_P$. Since $Q_1 \neq Q_2$, there exists some $q \in Q_1 \setminus Q_2$ or some $q \in Q_2 \setminus Q_1$. Without loss of generality, let $q \in Q_1 \setminus Q_2$. Then $\frac{q}{1} \in Q_1 R_P$ but $\frac{q}{1} \notin Q_2 R_P$, so $Q_1 R_P \neq Q_2 R_P$. Therefore, ϕ is injective. So ϕ is a bijection.

(2) Now consider $(\mathcal{P}, \subseteq)$ and $(\mathcal{P}_P, \subseteq)$ where $\mathcal{P}$ is the set of all prime ideals properly contained in P , and $\mathcal{P}_P$ is the set of all prime ideals in R_P . All primes in R must contain $\text{nil}(R)$ since R is commutative, and so by **Homework 4** $(\mathcal{P}, \subseteq) \subseteq ()$ the induced subposet obtained by removing $\hat{0} = \text{nil}(R)$, every proper ideal is contained in some maximal element, $\mathcal{P}$ is We proved that both of these are posets with $\hat{0}$. Well, if $Q_1 \subseteq$ □

Theorem 8. *If R is a Noetherian Domain and $P \subset R$ is a prime ideal minimal over $I_n = \langle p_1, \dots, p_n \rangle$ where $p_1, \dots, p_n$ are primes, then $\text{ht}(P) \leq n$.*

Proof. Recall that by **Lemma 4**, there exists a minimal prime ideal P over any ideal I in R . Let $I = \langle p_1, \dots, p_n \rangle$ where $p_1, \dots, p_n$ are primes, and let P be a minimal prime ideal over I . Now recall that the localization R_P of R at P defined via

$$R_P = \left\{ \frac{r}{q} : r \in R, q \in R \setminus P \right\} \text{ is a local ring with maximal ideal } PR_P \text{ by } \mathbf{Homework 3}.$$

If $n = 1$, $I = \langle p_1 \rangle$ and since p_1 is a prime element, I is a prime ideal, and $P = \langle p_1 \rangle$. By **Lemma 6**, $\text{ht}(P) \leq 1$.

Suppose the statement holds for some $k \geq 1$. Let $I = \langle p_1, \dots, p_k, p_{k+1} \rangle$ where $p_1, \dots, p_{k+1}$ are primes, and let P be a prime ideal minimal over I .

Consider any chain $Q_1 \subseteq Q_2 \subseteq \dots \subseteq Q_k = Q$ of prime ideals terminating at Q , and let $j \in [k]$ be maximal such that $Q_j \subset Q$. Then $Q_1 \subset \dots \subset Q_j \subset Q$, so $j \leq \text{ht}(Q) \leq k$.

Since $Q \subseteq P$, we may extend this chain by adding P at the end:

$$Q_1 \subset \dots \subset Q_j \subset Q \subseteq P.$$

Passing to the quotient R/Q , any strict containment between Q and P corresponds to a strict containment between 0 and P/Q . Since R/Q is a Noetherian domain and P/Q is a prime ideal minimal over $\langle [p_{k+1}]_Q \rangle$, Lemma 6 gives $\text{ht}(P/Q) \leq 1$. Hence there can be at most one strict containment between Q and P .

Therefore any chain of prime ideals terminating at P has length at most $j+1 \leq k+1$, so $\text{ht}(P) \leq k+1$. $\square$

Problem 1. Let R be a UFD, and let a and b be nonzero non-unit elements of R . Prove that a and b have a gcd, and describe it in terms of their prime factorizations.

Proof. Since R is a UFD, there exist irreducible decompositions $a = a_1 \cdots a_n, b = b_1 \cdots b_m$. Recall that any irreducibles in a UFD are also prime elements, so these are also prime products. Now let $G = \{p_1, \dots, p_N\}$ be some minimal collection of non-associate primes appearing in both a and b , which must exist since both have finite decompositions. Minimal here means we can express both a and b as a product within G . Then there exists α_i and β_i such that $a = \prod_{i=1}^N g_i^{\alpha_i}$ and $b = \prod_{i=1}^N g_i^{\beta_i}$.

Now let $d = \prod_{i=1}^N g_i^{\gamma_i}$ where $\gamma_i = \min\{\alpha_i, \beta_i\}$ for each $i \in [N]$. Behold, $\gamma_i \leq \alpha_i, \beta_i$ and so

$$\begin{aligned} g \prod_{i=1}^N g_i^{\alpha_i - \gamma_i} &= \left(\prod_{i=1}^N g_i^{\alpha_i} \right) \left(\prod_{i=1}^N g_i^{\alpha_i - \gamma_i} \right) = a \implies g \mid a \\ g \prod_{i=1}^N g_i^{\beta_i - \gamma_i} &= \left(\prod_{i=1}^N g_i^{\beta_i} \right) \left(\prod_{i=1}^N g_i^{\beta_i - \gamma_i} \right) = b \implies g \mid b \end{aligned}$$

Now consider any common divisor d of a and b . By **Lemma 1**, d can be expressed as some product of primes appearing in a and b and some units. Then obviously it is expressible as some unit multiplied by some product in G . That is, $d = u \prod_{i=1}^N g_i^{\delta_i}$ for some unit $u \in R$ and $\delta_i \leq \alpha_i, \beta_i$ for each $i \in [N]$ since these are non associate primes. Well, $\gamma_i = \min\{\alpha_i, \beta_i\} \in \{\alpha_i, \beta_i\}$ for each $i \in [N]$, so $\delta_i \leq \alpha_i, \beta_i \implies \delta_i \leq \gamma_i, \forall i \in [N]$. Therefore,

$$d \prod_{i=1}^N g_i^{\gamma_i - \delta_i} = \left(\prod_{i=1}^N g_i^{\delta_i} \right) \left(\prod_{i=1}^N g_i^{\gamma_i - \delta_i} \right) = \prod_{i=1}^N g_i^{\gamma_i} = g \implies d \mid g.$$

Thus, g is a gcd of a and b .

□

Problem 2. Let R be a PID and let S be a multiplicative set of R , and let $P \subseteq R$ be a prime ideal.

- (a) Prove that R/P is a PID.
- (b) Prove that $S^{-1}R$ is a PID.

Proof. **(a)** Consider any ideal $I_P \subseteq R/P$ and let $\pi : R \rightarrow R/P$ be the canonical homomorphic projection. Then the preimage $I = \pi^{-1}(I_P)$ of I_P is an ideal. Since R is a PID, $I = \langle i \rangle$ for some $i \in R$. Then $I_P = \pi(I) = \pi(\langle i \rangle) = \langle \pi(i) \rangle$, so I_P is principal.

(b) Consider any ideal $I_S \subseteq S^{-1}R$ and let $\iota : R \rightarrow S^{-1}R$ be the inclusion mapping defined by $\iota(r) = \frac{r}{1}$, which is clearly a homomorphism. Then the preimage $I = \iota^{-1}(I_S)$ of I_S is an ideal. Since R is a PID, $I = \langle i \rangle$ for some $i \in R$. Then $I_S = \iota(I) = \iota(\langle i \rangle) = \langle \iota(i) \rangle$, so I_S is principal. $\square$

Problem 3. Let R be a Noetherian UFD.

- (a) Prove that every descending chain of prime ideals

$$P_0 \supseteq P_1 \supseteq \cdots \supseteq P_n \supseteq \cdots$$

in R stabilizes.

- (b) The height of a prime ideal P is (if finite) the maximum length h of a chain of prime ideals

$$P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_h = P.$$

Prove that every prime ideal in R of height 1 is principal.

Proof. (b) This is a direct consequence of **Lemma 5**. If P has height 1, then P is principal.

(a)

Theorem 4. If R is a Noetherian UFD and $P = \langle p_1, \dots, p_n \rangle$ is a prime ideal generated by n prime elements, then $\text{ht}(P) \leq n$.

Proof. We prove the stronger statement that if $I = \langle p_1, \dots, p_n \rangle$ is an ideal generated by n prime elements and P is a prime ideal minimal over I , then $\text{ht}(P) \leq n$. The result follows immediately since if $P = \langle p_1, \dots, p_n \rangle$ is itself prime, then P is minimal over itself.

If $n = 1$, let P be prime minimal over $\langle p_1 \rangle$. Since p_1 is a prime element, $\langle p_1 \rangle$ is a prime ideal, so $P = \langle p_1 \rangle$. By Lemma 6, $\text{ht}(P) \leq 1$.

Suppose the statement holds for $n = k$. Let $I = \langle p_1, \dots, p_k, p_{k+1} \rangle$ and let P be a prime ideal minimal over I . Set $J = \langle p_1, \dots, p_k \rangle$. Since R is Noetherian, there exists a prime ideal Q minimal over J . By the induction hypothesis, $\text{ht}(Q) \leq k$.

Let $P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_m = P$ be a chain of prime ideals. Let i be the largest index such that $P_i \subseteq Q$. Then $P_0 \subsetneq \cdots \subsetneq P_i \subseteq Q$, so $i \leq \text{ht}(Q) \leq k$.

Since P is minimal over I , we have $I \subseteq P_{i+1}$. Thus $p_1, \dots, p_k \in P_{i+1}$, so $Q \subseteq P_{i+1}$. Passing to the quotient R/Q , the ideals $P_{i+1}/Q \subsetneq \cdots \subsetneq P_m/Q = P/Q$ form a chain of prime ideals. Since R/Q is a Noetherian domain and P/Q is a prime ideal minimal over $\langle [p_{k+1}]_Q \rangle$, Lemma 6 gives $\text{ht}(P/Q) \leq 1$. Hence $m - i \leq 1$.

Therefore $m = i + (m - i) \leq k + 1$, so $\text{ht}(P) \leq k + 1$. This completes the induction. □

□

Problem 5. Consider the ring

$$R := \mathbb{Z}[\sqrt{-5}] := \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}.$$

- (a) Prove that $R \cong \mathbb{Z}[x]/(x^2 + 5)$.
- (b) Since R is a subring of $\mathbb{Q}(\sqrt{d})$ where $d = -5$, the norm N defined in Homework 2 will still have the property that

$$N(zw) = N(z)N(w)$$

in R . Use this to prove that the units in R are ± 1 .

- (c) Prove that 2, 3, $(1 + \sqrt{-5})$, and $(1 - \sqrt{-5})$ are all irreducible non-associate elements of R .
- (d) Prove that R is not a UFD.

Problem 6. Let R be a Euclidean domain which is not a field. Prove that there exists a nonzero, non-unit element $c \in R$ such that for all $a \in R$, there is some $q, r \in R$ such that

$$a = qc + r$$

and $r = 0$ or r is a unit.

Hint: Pick c so that $N(c)$ is a minimal positive value among norms of non-units.

Problem 7. Consider the ring

$$R := \mathbb{Z} \left[\frac{1 + \sqrt{-19}}{2} \right] := \left\{ a + b \frac{1 + \sqrt{-19}}{2} \mid a, b \in \mathbb{Z} \right\}.$$

- (a) Since R is a subring of $\mathbb{Q}(\sqrt{d})$ where $d = -19$, the norm N defined in Homework 2 will still have the property that

$$N(zw) = N(z)N(w)$$

in R . Prove that the smallest values of $N(z)$ for $z \in R$ are 0, 1, 4, 5 and that

$$N\left(a + b \frac{1 + \sqrt{-19}}{2}\right) \geq 5 \quad \text{if } b \neq 0.$$

- (b) Prove that the units in R are ± 1 .
- (c) Prove that if $c \in R$ satisfies the condition in the previous exercise, then c must divide 2 or 3 in R , and conclude $c = \pm 2$ or ± 3 .
- (d) Show that there is no q, r such that

$$\frac{1 + \sqrt{-19}}{2} = qc + r$$

with $c = \pm 2, \pm 3$ and $r \in \{0, \pm 1\}$, and conclude that R is not a Euclidean domain.